

Lab 01: Half-Wave and Full-Wave Rectification

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ECE 3201 Electronic Circuit Design and Analysis

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1 Abstract

This lab introduced the concept of rectification using semiconductor diodes as nonlinear circuit elements. Half-wave and full-wave rectifiers were constructed, measured, and analyzed to demonstrate the conversion of alternating current (AC) into direct current (DC). The effect of diode forward voltage drop and peak inverse voltage (PIV) were considered, and ripple reduction techniques were explored using filter capacitors. Experimental results were compared to theoretical predictions, and discrepancies were explained in terms of practical component tolerances and non-ideal diode behavior. The lab provided a foundational understanding of rectifier operation, efficiency, and filtering, which are essential for power electronics applications.

2 Introduction

A diode is a two-terminal nonlinear device that ideally conducts with negligible drop when forward-biased and blocks when reverse-biased. Real silicon diodes exhibit a forward drop near 0.7 V and finite reverse leakage, which directly impact rectifier performance. These one-way conduction properties (mechanically analogous to a check valve) enable conversion of AC to unidirectional DC. In this lab we use half-wave, bridge, and center-tap full-wave rectifiers to quantify the effects of diode forward drop, PIV, and RC filtering on average output and ripple.



Figure 1: The Schematic Diagram of a Diode

Think of the diodes mechanical analog being the check-valve. A check-valve in a fluid system allows the fluid to flow in one direction only, with a mechanical gate swinging down preventing the backflow of fluid in the improper direction. The diode acts very much the same way. These characteristics can be exploited to produce circuitry that was previously not possible with the components of ECE 2001 such as rectification circuits, voltage-shifting circuits, and voltage-limiting currents.

3 Objective

The objective of this lab is to investigate the operation of half-wave and full-wave rectifier circuits using diodes, and to analyze their ability to convert alternating current (AC) signals into direct current (DC). We will construct and measure different rectifier configurations, compare theoretical and experimental results for output voltage and ripple, and evaluate the effects of circuit parameters such as diode forward voltage drop, peak inverse voltage (PIV), and filter capacitors on rectifier performance.

4 Theory

Rectification is the process of converting an alternating current (AC) signal into a direct current (DC) signal using semiconductor diodes. A diode is a nonlinear device that ideally behaves as a short circuit when forward biased and as an open circuit when reverse biased. In practice, a forward voltage drop of approximately 0.7 V for silicon diodes must be considered.

4.1 Half-Wave Rectifier

In a half-wave rectifier, the diode conducts only during the positive half-cycle of the input sinusoidal signal. During the negative half-cycle, the diode is reverse biased, and the output

voltage is zero. For an ideal diode, the instantaneous output is

$$v_o(t) = v_i(t), \quad 0 < \omega t < \pi,$$

and

$$v_o(t) = 0, \quad \pi < \omega t < 2\pi.$$

Accounting for the diode forward voltage drop V_D , the output becomes

$$v_o(t) = v_i(t) - V_D, \quad v_i(t) > V_D.$$

The average (DC) output voltage is approximated as

$$V_{DC} \approx \frac{V_m}{\pi} - \frac{V_D}{2},$$

where V_m is the peak input voltage. The main limitation of the half-wave rectifier is its low efficiency since only half of the input waveform contributes to the output.

4.2 Full-Wave Rectifier

In a full-wave rectifier, both halves of the input waveform are utilized. This can be achieved using either a center-tapped transformer with two diodes or a four-diode bridge rectifier. In both cases, the output waveform is unidirectional, effectively doubling the frequency of the rectified signal. The average output voltage for an ideal diode is

$$V_{DC} \approx \frac{2V_m}{\pi}.$$

Compared to the half-wave rectifier, the full-wave rectifier provides a higher average DC voltage and improved efficiency.

4.3 Filtering

The output of a rectifier is not a constant DC value but contains ripples. To reduce ripple voltage, a filter capacitor is connected across the load resistor. The capacitor charges quickly to the peak value of the rectified waveform and discharges slowly through the load during the non-conducting period of the diode. The resulting ripple voltage can be approximated as

$$V_r \approx \frac{V_p}{fRC},$$

where V_p is the peak output voltage, f is the frequency of the rectified waveform, R is the load resistance, and C is the filter capacitance. For effective filtering, the condition $RC \gg T$ (where $T = 1/f$) must be satisfied.

4.4 Peak Inverse Voltage (PIV)

The peak inverse voltage is the maximum reverse bias voltage that a diode must withstand during operation. In a half-wave rectifier, the PIV is equal to the peak input voltage V_m . In a bridge rectifier, each diode experiences a PIV of V_m , whereas in a center-tap full-wave rectifier, the PIV is $2V_m$. Thus, diode selection must ensure that the breakdown voltage rating exceeds the PIV by a safe margin.

5 Experimental Procedures

5.1 Circuit One

The first circuit analyzed was a simple half-wave rectifier using a silicon diode outlined in Figure 2. This involved an input signal of $10V_{pp}$ at 1KHz from the signal generator and a 1N457 diode.

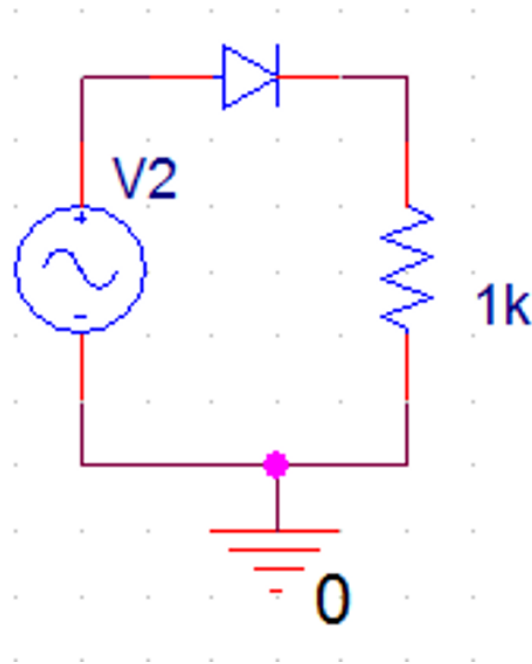


Figure 2: A Half-Wave Rectifier Circuit

5.2 Circuit Two

The second circuit analyzed was a Full-Wave Rectifier and involved a slightly more complicated test procedure. The full wave rectifier circuit also utilized the 1N457 diode, but this time 4 were used in order to obtain the correct characteristics of a full-wave rectification circuit. This also employed the use of a transformer circuit that was tuned to obtain $120V_{rms}$ from the wall socket at 60Hz and transform that down via a secondary winding to an obtained voltage of $16.0V_{rms}$. The circuit schematic is shown in Figure 3

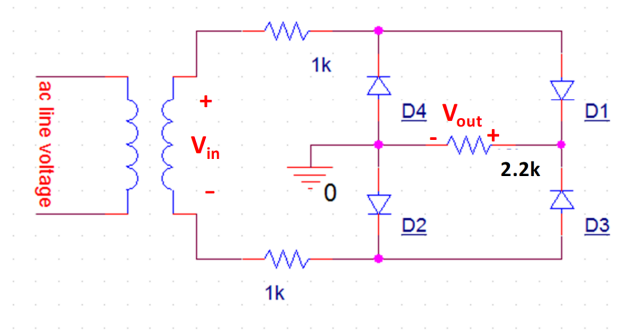


Figure 3: A Full Bridge Rectifier Circuit

5.3 Circuit Three

The third circuit involved a more simplified paradigm of the Full Wave rectification circuit, this time only employing one half of the 1N457 diode.

The same circuit specifications were used, but this time the center tap of the transformer was regarded as reference. This effectively splits the voltage felt at the terminals in half as shown in the following Results section. The circuit is shown in Figure 4.

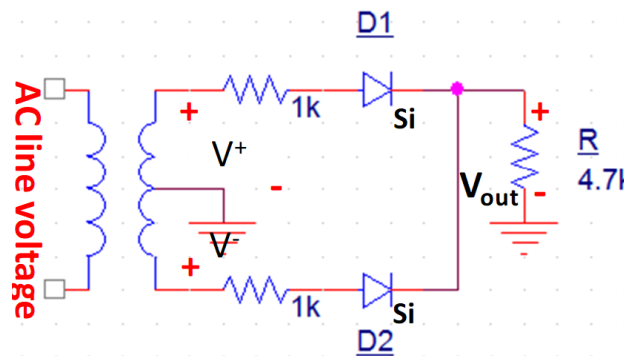


Figure 4: A Full Wave Center Tap Configured Circuit

5.4 Circuit Four

The final circuit involved a design skill and the use of some formulas that were previously mentioned in the Theory section. Using a target ripple voltage of 0.8V with an input of a $5V_{pp}$ sine wave at 60Hz, a resistance and capacitance value had to be chosen such that these design conditions hold true. The circuit is shown in Figure 5.

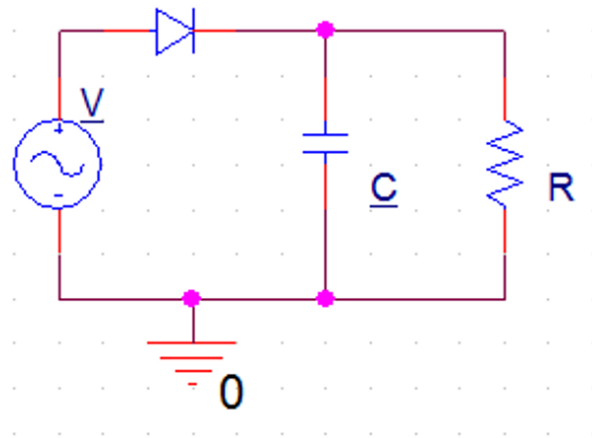


Figure 5: Rectifier Circuit with Filtering Capacitor

6 Results

6.1 Circuit One

The input signal can be seen in Figure 6 and the respective output signal can be seen in Figure 7. The behavior follows expectations with the diode obtaining some amount of voltage during the drop. Notice the peak input voltage tops out at 4.793V, while the peak output voltage tops out at 3.914V.

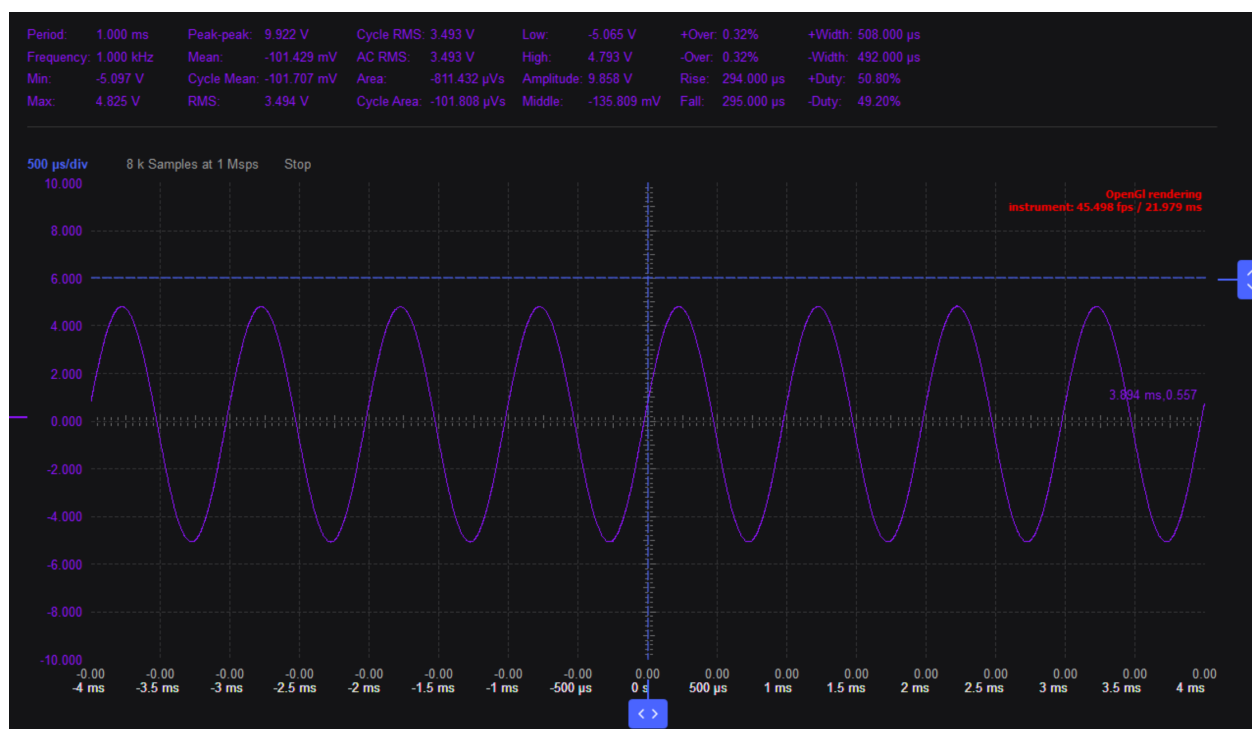


Figure 6: Input Waveform to Circuit One



Figure 7: Output Waveform to Circuit One

Subtracting the two and accounting for the DC offset in the output gives

$$4.793 \text{ V} - (3.914 \text{ V} + 0.1598 \text{ V}) = 0.7192 \text{ V}.$$

This aligns with the expected forward drop plus loading effects.

Taking the RMS voltage off of the resistor, the following was obtained:

$$V_{R_{rms}} \text{ Multimeter} = 1.530 V_{rms}$$

$$V_{R_{rms}} \text{ Oscilloscope} = 1.878 V_{rms}$$

The oscilloscope reports *true* RMS (DC + AC):

$$V_{\text{RMS,scope}} = 1.878 \text{ V}.$$

The multimeter reports the *AC-only* RMS, these are related by

$$V_{\text{RMS,scope}}^2 = V_{\text{DC,meas}}^2 + V_{\text{RMS,AC}}^2 \implies V_{\text{RMS,AC}} = \sqrt{V_{\text{RMS,scope}}^2 - V_{\text{DC,meas}}^2}.$$

Substituting the measurements,

$$V_{\text{RMS,AC}} \approx \sqrt{(1.878)^2 - (1.055)^2} \approx 1.55 \text{ V},$$

which agrees closely with the multimeter reading

$$V_{R,\text{RMS}}(\text{multimeter, AC}) = 1.530 \text{ V}.$$

For reference, the ideal half-wave total RMS is $V_{p,\text{out}}/2 \approx 3.914/2 = 1.957$ V, reasonably close to the scope's 1.878 V given non-idealities.

For an *ideal* half-wave rectifier, $V_{DC} \approx V_{p,\text{out}}/\pi$, so

$$V_{DC,\text{ideal}} \approx \frac{3.914}{\pi} \approx 1.246 \text{ V}.$$

The measured mean was 1.055 V, lower as expected due to the diode drop and the slightly reduced conduction angle (the diode only conducts while $v_i(t) > V_D$).

The 1N4148 diode is a completely different diode than the one used in this experiment. Looking at the data sheet, the 1N4148 has a reverse breakdown voltage of 100V. The Peak Inverse Voltage for this circuit, or PIV, is simply 5V, which is 5% of this rating. Therefore, it heavily exceeds this requirement.

Upon decreasing the amplitude to $2V_{pp}$, the following differences are shown:



Figure 8: Output Waveform to Circuit One @ $2V_{pp}$

VR @2V-measured becomes 58.819mV and the calculated VR becomes 107.6mV . Because V_D is a small fraction of the peak ($0.7/5 \approx 14\%$), the ideal formula $V_{DC} \approx V_p/\pi$ remains accurate and the conduction angle is near π . At $2 V_{pp}$ ($V_p = 1$ V), V_D/V_p is large ($\approx 70\%$), the conduction angle shrinks, and non-idealities (diode curve, meter resolution, waveform distortion) dominate, pulling the measured DC below the ideal estimate.

6.2 Circuit Two

Circuit two, shown in Figure 3 involved taking measurements of the output resistor again, however the characteristics now show a full wave rectification circuit output. First, the input voltage shown in 9 shows a $16V_{rms}$ reading off of the transformer windings provided by the lab.

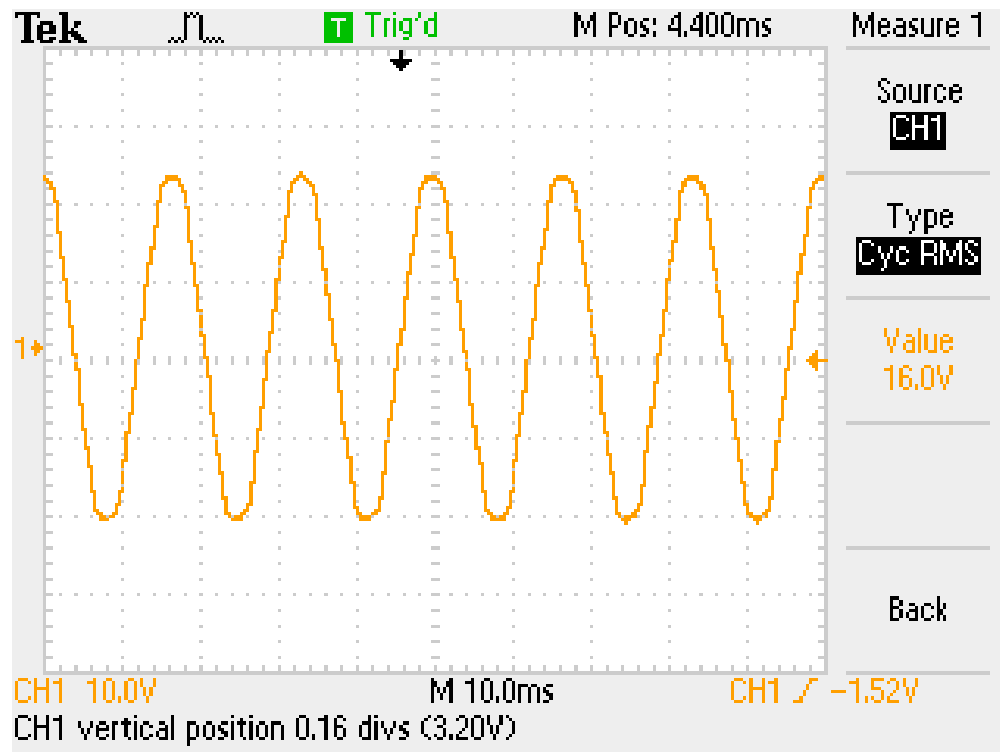


Figure 9: Input Waveform to Circuit Two

The output, as measured from the resistor displayed both sides of the input sine wave, but on the positive side. As there are four diodes now in a bridge configuration, half of the diodes ‘fire’ when the input sine wave is in the positive half of conduction, and the other half ‘fire’ when the signal is in the negative half. This allows for the entire signal to pass through as positive and for better circuit efficiency at the cost of the additional components needed to build the circuit. The Mean and Max outputs of these signals are shown in Figures 10 and 11 respectively.

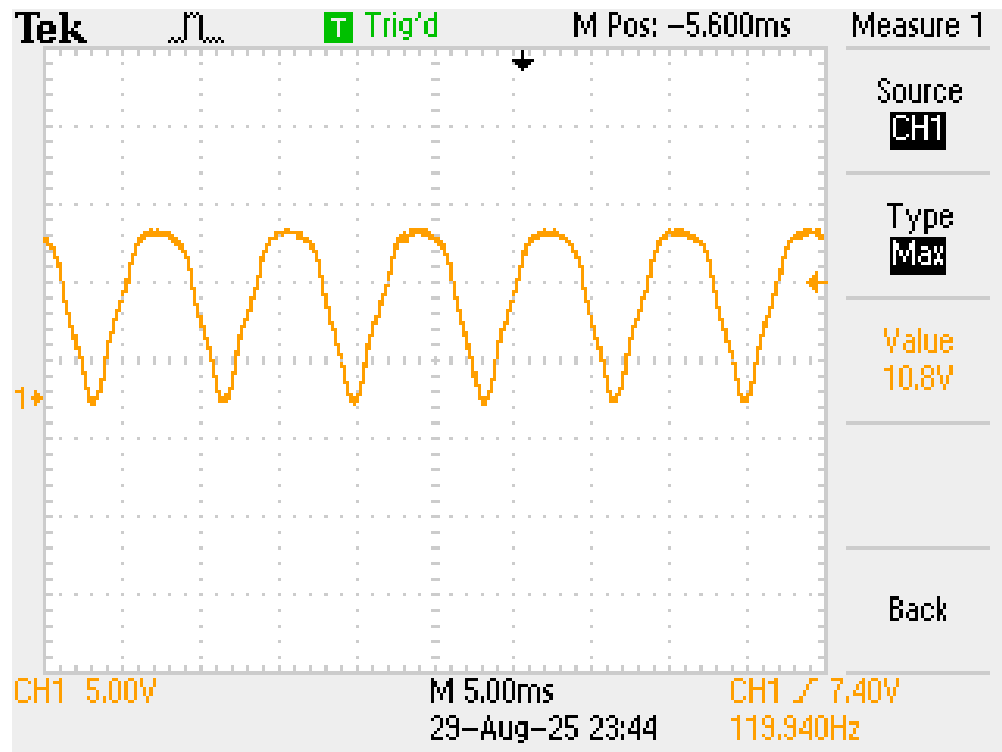


Figure 10: Output Max Waveform of Circuit Two

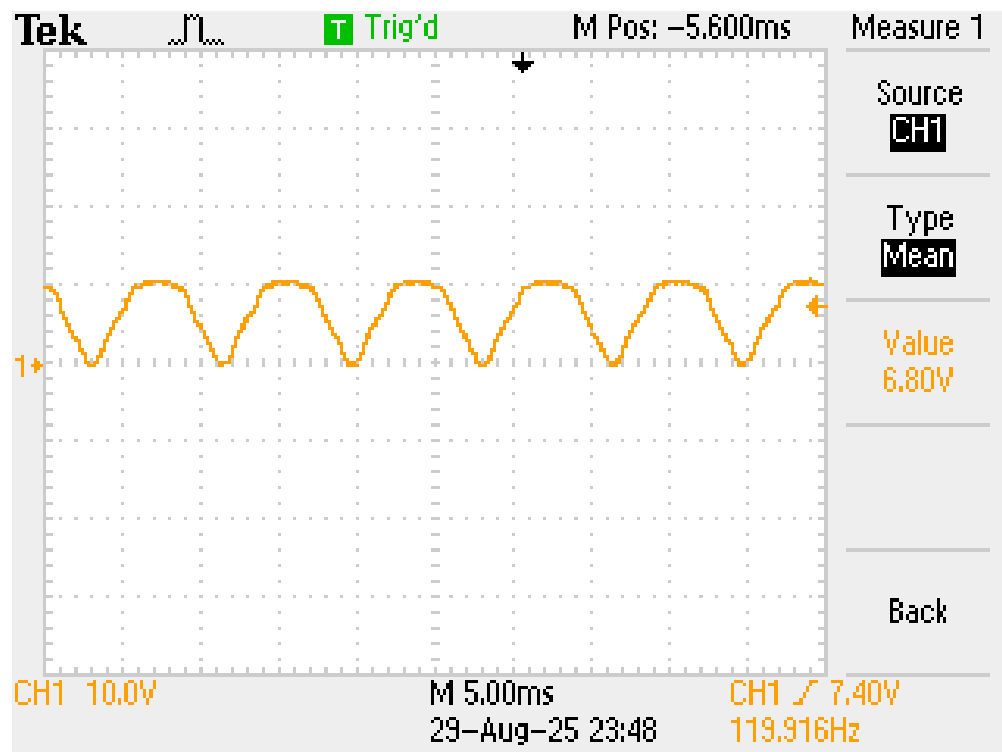


Figure 11: Output Mean Waveform to Circuit Two

This resulted in the following values:

$$V_{IN} = 15.9V$$

$$V_{OUT_{MAX}} = 10.8V$$

$$V_{OUT_{DC}} = 6.8V$$

Intuitively, these all make sense. On each half of the signal (positive and negative) there are now two diodes which drop the voltage when they are forward biased. Setting these to approximately 0.7V mentally, the max peak voltage alongside the added resistance from the resistor make the max voltage as expected. The DC Component of the circuit is also closer to the average as each period now has both halves of the sine wave included, allowing use of the mean measurement of the oscilloscope to be accurate compared to before in Circuit One.

6.3 Circuit Three

Circuit Three introduced the use of the center tap of the transformer. The center tap effectively cuts the transformer into two halves right down the center.

This is shown in the input waveforms. The positive half is shown in Figure 12 and the negative half is shown in Figure 13.

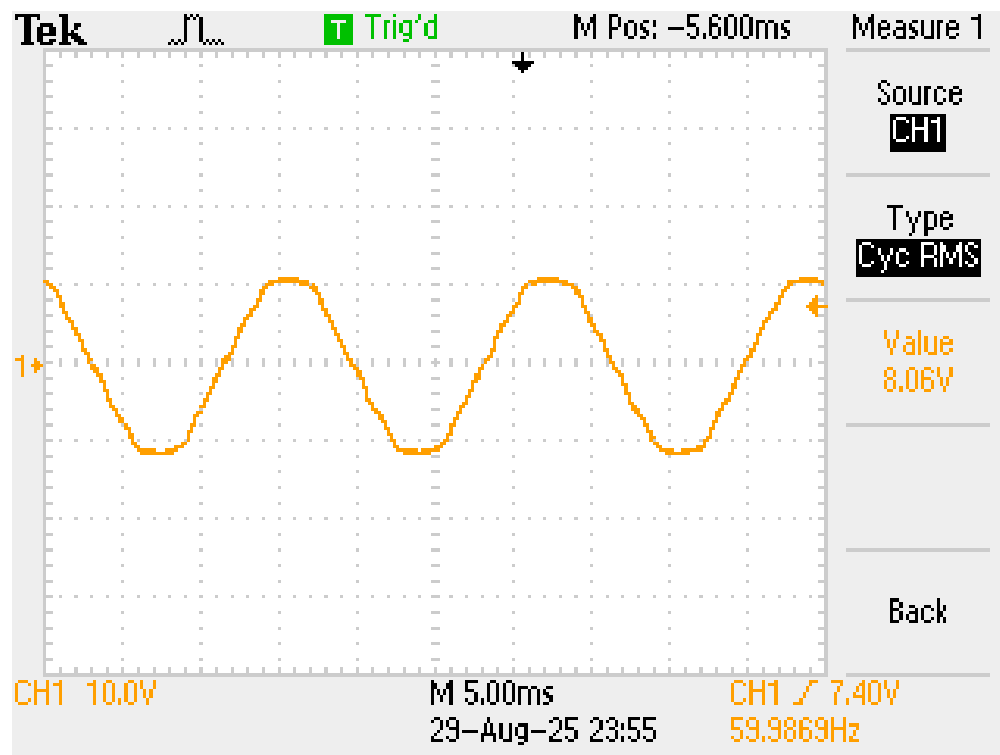


Figure 12: Positive Transformer Half Side Input to Circuit 3

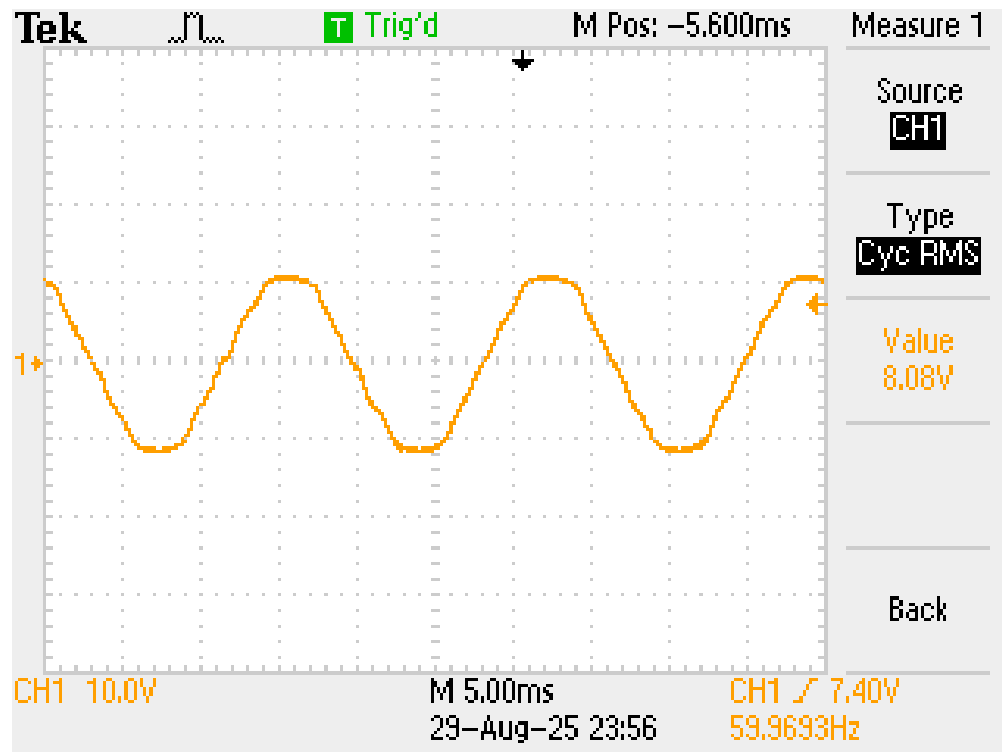


Figure 13: Negative Transformer Half Side Input to Circuit 3

This results in:

$$V_{IN+} = 8.06V_{RMS}$$

$$V_{IN-} = 8.08V_{RMS}$$

$$V_R = 6.03V_{RMS}$$

The output waveform can be seen in Figure 14.

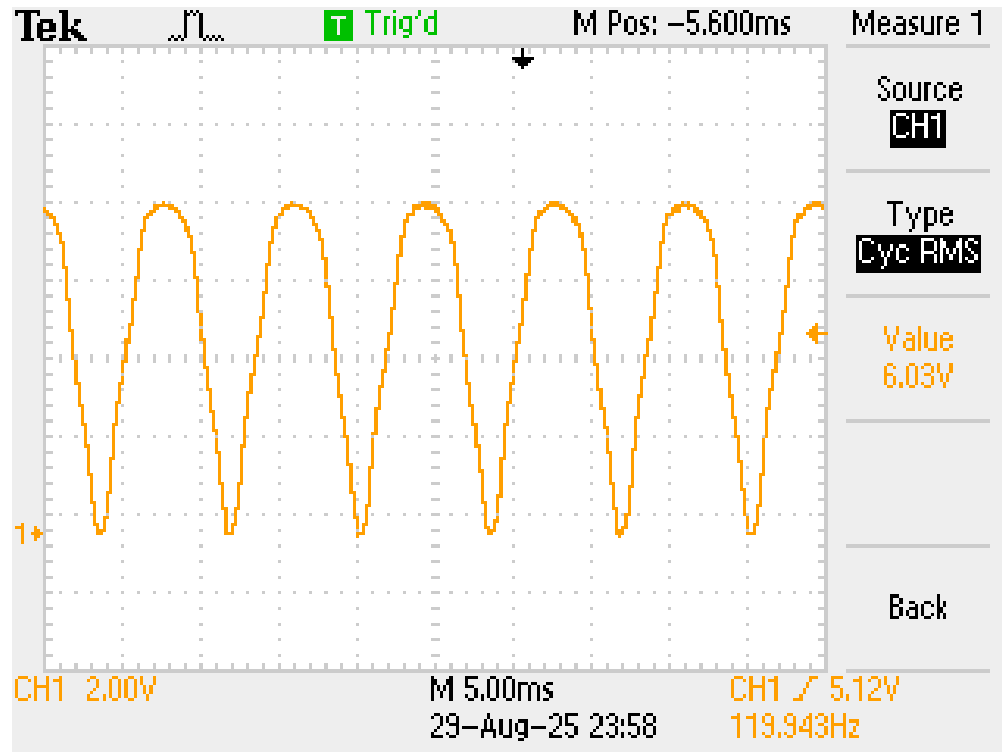


Figure 14: Circuit Three Output Waveform

Which is very similar indeed to Circuit Two. One could say that the outcome is the same, but with different physical characteristics and costs associated with each circuit. I'll delve into some details here.

The output, as taken from the resistor shows a response similar to the one in Circuit Two, but with the PIV doubled. This center-tapped full-wave rectifier behaves like a half-wave rectifier in the top half, and another one in the bottom half, phase shifted so that the output is a similarly clean, energetic output waveform.

In terms of the output voltage, since there is one less diode on each half of the input waveform, that means that there is only one (0.7V) drop in potential voltage on each half of conduction compared to Circuit Two.

$$V_{out, \text{center-tap}} \approx V_m - V_D$$

$$V_{out, \text{bridge}} \approx V_m - 2V_D$$

To explain further on the aspect of the PIV calculation, consider:

$$PIV = 2V_S - V_D$$

compared to Circuit Two's:

$$PIV = V_S - 2V_D + V_d = V_S - V_D$$

The PIV is about half the value for the full-wave rectifier with a center-tapped transformer!

6.4 Circuit 4

For the rectifier with filter capacitor shown in Fig. 5, the ripple voltage is approximated by

$$V_r \approx \frac{V_p}{fRC},$$

where V_p is the peak input voltage, f is the frequency, R is the load resistance, and C is the filter capacitance.

Given:

$$V_p = \frac{5 V_{pp}}{2} = 2.5 \text{ V}, \quad f = 60 \text{ Hz}, \quad V_r = 0.8 \text{ V}, \\ R = 2.2 \text{ k}\Omega, \quad C = 22 \mu\text{F}.$$

Substituting into the ripple equation:

$$V_r = \frac{2.5}{60 \times 2200 \times 22 \times 10^{-6}}, \\ V_r = \frac{2.5}{2.904} \approx 0.86 \text{ V}.$$

Thus, the calculated ripple voltage with the chosen values is

$$V_r \approx 0.86 \text{ V},$$

which is close to the design target of 0.8 V.

Using the oscilloscope, the measured ripple voltage was found to be

$$V_{r,\text{measured}} \approx 0.96 \text{ V}.$$

as shown in Figure 15.

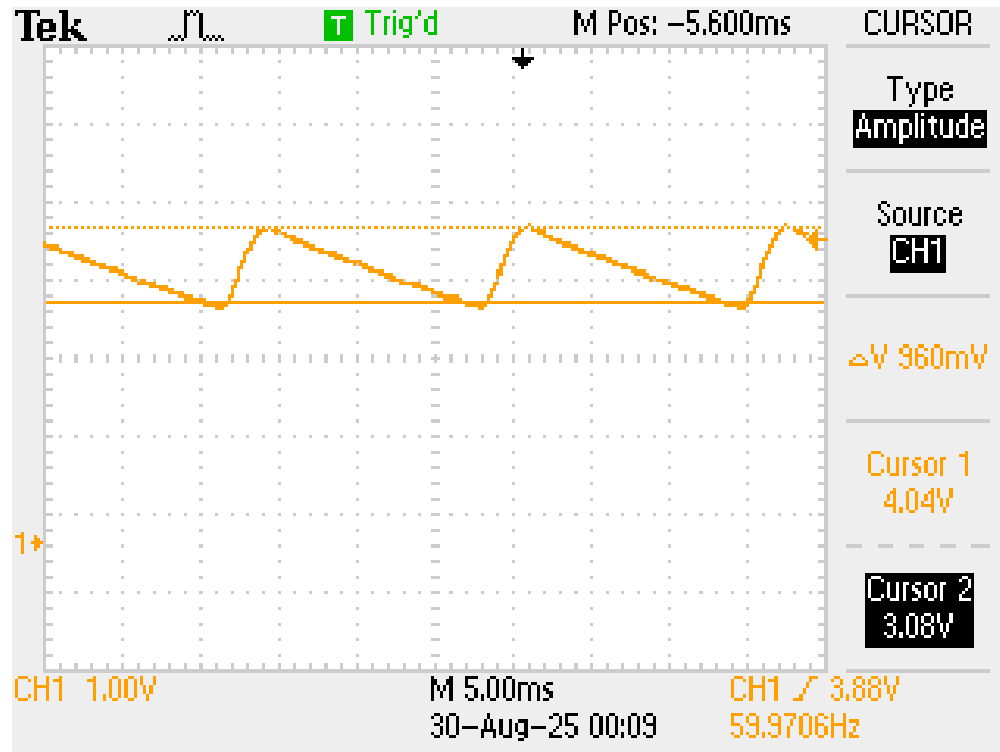


Figure 15: The Designed Circuit Ripple Voltage

The discrepancy between calculated and measured values is within an acceptable margin of error. It arises primarily from component tolerances (the measured resistance was $2.213\text{ k}\Omega$ instead of the calculated $2.367\text{ k}\Omega$), as well as diode forward voltage drops and non-ideal capacitor behavior. The slightly lower resistance reduced the RC time constant, allowing more discharge between cycles and producing a larger ripple voltage.

The condition $CR \gg T$ is essential for effective filtering. Here, CR is the discharge time constant of the capacitor through the load resistor, and $T = 1/f$ is the period of the rectified waveform. When $CR \gg T$, the capacitor discharges very little between successive peaks, minimizing ripple and yielding a nearly constant DC output. If this condition is not satisfied, significant ripple voltage appears, reducing rectification performance. Additionally, assuming $CR \gg T$ simplifies analysis by allowing exponential discharge terms to be approximated linearly, making ripple voltage estimation more straightforward.

7 Conclusion

The experiments validated core rectification principles and clarified how non-idealities shape real outputs. The half-wave rectifier delivered a low average DC level with substantial ripple, as expected when only one half-cycle contributes. Full-wave rectification improved efficiency by utilizing both half-cycles; the bridge topology incurred two forward drops per conduction path, while the center-tap topology incurred one but required roughly twice the PIV per diode.

The filtered design exercise emphasized the importance of the $CR \gg T$ condition. With

$R = 2.2\text{ k}\Omega$ and $C = 22\text{ }\mu\text{F}$, the predicted ripple ($\approx 0.86\text{ V}$) was close to the target 0.8 V and to the measured 0.96 V . The residual difference is consistent with component tolerances, diode forward drop variation, and capacitor ESR/ESL. Overall, measurements tracked first-order theory well, providing a reliable foundation for designing rectifier stages and for extending to regulated supplies where ripple and PIV margins are critical.